SAFESIGHT | CIS-HW-01

Edge AI Safety Monitoring

A readable guide to the project idea, design, evidence, current status, and pitch preparation.

Status as of 05 November 2026
Software prototype in progress. Final model training, evaluation, real camera testing, and Jetson validation remain open.

In one sentence

SafeSight is a fixed-camera prototype that combines visual cues about personal protective equipment (PPE), restricted-zone activity, and fire/smoke with readable risk rules, then presents alerts and event context in a local dashboard.

How to use this guide
Read the design and risk sections to understand the concept. Use the progress, evaluation, and limitations sections to answer questions accurately. The final pages include a short speaking outline and likely reviewer questions. This document is context, not proof of deployment readiness.

1. Purpose and scope

The project explores whether an affordable edge-computing setup can help a small team notice and review selected safety conditions in a fixed area. A camera supplies video. Computer-vision models estimate people, PPE, and fire/smoke cues. Zone geometry tracks whether people enter a configured area. A fuzzy rule system combines cues into a 0-100 score and a Low, Medium, High, or Critical band.

The intended operator view shows the current band, contributing inputs, recent alert events, timestamps, and saved snapshots. Processing is designed to run locally at the edge. The current laptop implementation is a development prototype, not proof of Jetson Nano performance.

Initial use case

- A fixed camera observes one known view; a manually configured polygon marks a restricted or hazardous region.
- The software raises a review signal when selected visual indicators combine into a higher band.
- An operator checks the video/event context and follows the site's existing procedures.

What is not claimed

- This does not replace PPE policy, human supervision, emergency procedures, or certified fire alarms.
- It cannot guarantee every person, helmet, vest, flame, or smoke plume will be detected.
- The score is not the probability of an accident; it is a rule-based prioritization score.
- No accuracy, site generalization, or Jetson real-time performance claim exists yet.

What success would mean

A credible pilot would show the full workflow on the intended camera, report held-out detector measurements, measure end-to-end speed, and obtain safety-professional review of the risk scenarios. It would remain a decision-support prototype.

2. System architecture

The live data path is a chain. Each stage adds context and passes a small set of values onward.

Stage	What it does	Current caveat
1. Video	Reads camera frames or recorded clips.	An actual event-camera run is still needed.
2. Vision	Detects people, head, helmet, vest, fire, and smoke.	Final trained models and held-out results are not available.
3. Context	Tracks people, estimates feet, checks zone entry/time, associates PPE cues.	Real-view zone and occlusion checks remain.
4. Risk	Combines normalized signals with fuzzy rules and explicit overrides.	Independent expert agreement is unmeasured.
5. Response	Shows dashboard state, alert output, event log, and snapshot.	Live full pipeline and physical GPIO are pending.

Key mechanics

- A person's bottom-centre box point approximates their feet. If it falls inside the polygon, that track is counted in the zone.

- Tracking maintains an intrusion timer while a person remains inside; leaving or disappearing from detections resets the timer.

- The PPE association code uses box overlap and body-region heuristics to associate helmets/heads and vests to a person. Occlusion and overlapping people can still cause errors.

- PPE detection runs every frame; fire/smoke checks are scheduled less often (first frame, then every tenth frame) and recent confidence is smoothed.

Edge-first goal

Process video on or near the camera location, avoiding a cloud dependency for the core workflow. The laptop prototype has not yet been benchmarked on the target board.

3. Risk score explained

A two-stage Mamdani fuzzy system handles inputs that can be uncertain or borderline. Inputs receive low, medium, or high membership. Readable rules combine them into an intermediate hazard estimate and then a risk band.

Approved input formulas

Input	Live pipeline formula	Meaning
PPE load	min(1, (2 x missing-helmet people + missing-vest people) / (3 x people)); empty scene = 0.	Higher means more missing-PPE evidence; helmet evidence is weighted twice vest evidence.
Zone pressure	0.5 x min(intruders / 2, 1) + 0.5 x min(longest intrusion seconds / 5, 1); no intruders = 0.	Combines crowd size and duration.
Proximity	1 minus the zone monitor safe-margin value, normalized 0-1.	Higher means closer to or inside the hazard boundary.
Fire evidence	Maximum fire/smoke confidence from the latest ten checks.	Higher means stronger recent model evidence.

Rules and output

- Stage 1 has 27 combinations of low/medium/high PPE, zone, and proximity inputs.

- Stage 2 has nine combinations of hazard and fire evidence.

- The engine uses triangular membership functions, min for AND, max aggregation, and centroid defuzzification.

- A monotonic calibration corrects small decreasing regions in the raw centroid surfaces. Runtime output is calibrated interpolation, not raw centroid at every point.

Condition	Result
Fire evidence >= 0.7	Score 100, Critical.
Observed unhelmeted intrusion >= 3 sec	At least 65, High.
Observed unhelmeted intrusion >= 8 sec	At least 85, Critical.
Bands	0-25 Low; above 25-50 Medium; above 50-75 High; above 75-100 Critical.

Software checks cover 27,300 monotonic grid edges and 10,000 continuous increase comparisons. These checks confirm code behavior; they are not detector accuracy or expert validation.

4. Data and model plan

Public datasets were used because photos from the eventual demo environment are unavailable. The baseline helps build the pipeline, but does not establish performance in a different room, with a different camera, light, background, or PPE.

Dataset	Prepared data	Use and caution
SH17 PPE	8,099 image/label pairs. Train 5,669; validation 1,214; test 1,216.	Classes: person, head, helmet, vest. Seed-42 split; exact image hashes checked. Public-only baseline.
D-Fire	21,501 retained images: 14,109 train; 3,098 validation; 4,294 test.	Fire and smoke; upstream splits preserved. 26 invalid pairs were excluded and recorded.

Training approach

- PPE starts from general COCO-pretrained YOLOv8n, reducing known SH17-specific test exposure.
- Train on train data; use validation for checkpoint selection; reserve the frozen test split for final evaluation only.
- Resume PPE, finish training, evaluate the frozen test once; then train and evaluate fire/smoke.
- Report per-class precision, recall, mAP, confusion matrices, and weak spots only after those evaluations exist.

PPE status as checked 05 Nov 2026

The latest saved checkpoint reports 26 completed epochs; the log shows epoch 27 had started before it stopped. No training process was active in the local process check. The checkpoint can be resumed, but there is no completed final model or frozen-test result.

Fire status

Dataset preparation is complete. Fire-model training and held-out test evaluation have not started.

Dataset counts are not accuracy. Exact byte-duplicate checks do not rule out near-duplicate frames, source overlap, or domain shift. SH17-derived weights are subject to CC BY-NC-SA 4.0 terms. Check D-Fire upstream terms before redistributing data or weights.

5. Dashboard and alert workflow

The dashboard is intended to show a camera view, risk band and score, input contributions, people and zone counts, and recent events with snapshot links. A CSV log provides a simple event record.

Alert behavior implemented in software

- The displayed risk can change immediately. Sound/buzzer event confirmation requires one continuous second at High or Critical.

- A confirmed Critical alert stays latched until the score remains below High for three continuous seconds.

- One event is logged per confirmed alarm episode and one additional event for later Critical escalation.

- Missing, stale, or failed inputs should show unavailable state rather than implying a safe Low result.

- Jetson.GPIO path targets BOARD pins 12 and 16 for buzzer and LED; only mock checks exist, no physical wiring test.

What the current preview proves

The responsive dashboard has been reviewed at desktop and mobile sizes. The scripted demo shows how score, reasons, events, and snapshots could appear. It uses no camera and no AI detections. Displayed score, FPS, and latency are illustrative and must not be quoted as model performance.

For a live demo it still needs

- Both final models, connected camera, and a zone polygon drawn on that exact camera view.

- A live walk-through to check detections, tracking, intrusion timers, alerts, and sound/output.

- Jetson Nano or target device, correctly wired GPIO hardware, and measurements on that device.

> **Zone caution**
> The polygon belongs to its camera view. Redraw and walk-test it after camera movement or a change in the demo scene.

6. Completed work and open gates

The following describes recorded software status, not a finished deployment.

Area	Status	Evidence / remaining gate
Setup and inference	Implemented	Supports webcam, video, image, image-folder inputs. Sample clip and image-folder checks were done; live webcam observation is open.
Data preparation	Implemented	Public SH17 and D-Fire prepared; labels, split isolation, frozen copies and hashes checked.
PPE training	Stopped before completion	Resume checkpoint at epoch 26; resume, finish, and evaluate frozen set.
Fire model	Not started	Training queue is defined; train after PPE, then evaluate frozen set.
Zones	Implemented; site check open	Polygon editor, tracking, timers and crowd logic have software checks; actual camera walk-through is open.
Fuzzy engine	Implemented; logic checked	27 + 9 rules and monotonic checks. Expert agreement not measured.
Dashboard	Implemented; simulated preview	31 tests and desktop/mobile browser checks recorded. Real-model live run remains open.
Phase 8 metrics	Tools ready; values pending	Needs final models, 60 seconds of recorded video with matching zone, and expert scenario labels.
Jetson integration	Pending hardware	Physical camera, Nano, wiring, GPIO, and performance are unverified.

Independent expert review

A project discussion considered arranging an expert review, but no 30 expert-labelled scenarios or agreement score are present in the project evidence. Do not claim expert validation until the labels and results exist.

7. Evaluation plan and claim discipline

Question	Evidence needed	What it can support
PPE model performance	One final evaluation on frozen PPE test; per-class precision/recall, mAP@0.50 and mAP@0.50:0.95, confusion matrix.	Performance on this public split only, with pretraining and domain-shift caveats.
Fire/smoke model	One evaluation on frozen D-Fire test; per-class metrics and confusion matrix.	Performance on retained D-Fire test images only.
Pipeline speed	At least 60 seconds of recorded video, matching polygon, named machine; full-app FPS and risk latency.	Speed on that device/clip; not Jetson camera latency unless separately measured.
Risk policy agreement	30 scenarios labelled by a safety professional; calculate agreement and document disagreements.	Agreement on those cases only.
Hardware operation	Actual camera, Nano, GPIO wiring, zone, and walk-in/out check.	Bounded hardware demo, not safety certification.

Valid to report now

- Dataset counts and data-preparation checks.
- Fuzzy-rule coverage and monotonicity software checks.
- Dashboard interaction and scripted demo, clearly labelled simulated.

Not available now

- Final detector accuracy, real-time FPS/latency, expert agreement, or physical Jetson performance.

Claim discipline

Use words such as prototype, estimate, public-data baseline, simulated preview, and needs field validation. Avoid certified, proven accurate, real-time on Nano, or prevents accidents unless future evidence supports them.

8. Limits and responsible use

A camera sees only part of a scene. Detections can fail because of distance, occlusion, lighting, angle, unusual PPE, carried helmets, or overlapping people. PPE-to-person matching is geometric and may assign equipment to the wrong person. A visual fire/smoke model can miss a hazard or issue false alerts.

Risk	Why it matters	Mitigation
Public-to-site shift	Event space may differ from training images.	Do not claim site accuracy; collect permitted site data later and keep evaluation separate.
False/missed cue	A missing box is not proof that PPE is absent or a place is safe.	Human review, explicit unavailable state, existing procedures.
Zone calibration	Wrong polygon or moved camera corrupts distance/entry estimates.	Draw on actual view; walk-test; recheck after movement.
Fire alert limits	Model is not a certified fire alarm.	Keep approved alarms; call it an experimental visual alert layer.
Privacy	Video may identify people or capture activity.	Obtain location/participant permission; minimize retention; secure files; follow local policy. No privacy audit is complete.
Licensing	Data, weights, and upstream code have separate terms.	Retain attribution and check terms before reuse or redistribution.
Target hardware	Laptop speed may not fit Nano limits.	Measure on actual target hardware; document any runtime changes.

SafeSight should never be the only safety control. Workers and operators must continue to rely on established procedures and certified equipment.

9. Next month: plan to event day

Step	Work	Completion evidence
1. Resume PPE	Restore datasets if needed and resume from GitHub checkpoint.	Final best checkpoint; run completes.
2. Evaluate PPE	Verify frozen hashes; evaluate once.	Per-class report and confusion matrix.
3. Train/evaluate fire	Use the prepared dataset after PPE job ends.	Fire model and held-out report.
4. Measure pipeline	Use >=60 seconds of permitted video and its matching zone.	Machine-labelled FPS and latency report.
5. Review scenarios	Get a safety professional to label 30 realistic inputs.	Agreement plus disagreements documented.
6. Rehearse	Use actual camera if available; otherwise label preview synthetic.	All numbers and limitations stated correctly.

Inputs needed from school/project owner

- Confirm Jetson Nano, camera, buzzer/LED, GPIO and access to the intended filming location.
- Get location and participant permission for any video. Do not imply site data was collected if it was not.
- Arrange independent safety-professional scenario review if available; otherwise keep agreement as future work.
- Keep camera placement stable long enough to draw and walk-test the restricted zone.

10. Pitch notes and likely questions

Suggested 30-second opening

Opening

SafeSight is a fixed-camera edge-AI prototype for safety monitoring. It looks for selected PPE and fire/smoke cues, tracks whether people enter a marked restricted zone, and combines those signals using readable fuzzy rules. The dashboard shows a risk band and why it changed, while logging the event for a person to review. We have built the main software workflow and public-data preparation. Model training and real Jetson/camera validation are still in progress, so we present this as a prototype, not a proven safety product.

Questions you may be asked

Question	Honest response
Is it running on Jetson Nano?	Not yet. Laptop software is being developed; Nano performance and physical GPIO are unverified.
What accuracy do you have?	No final frozen-test PPE or fire results yet. We will report class metrics after training/evaluation.
Does the dashboard prove AI works?	No. The current dashboard preview is a scripted simulation, not camera detections.
Why fuzzy logic?	It combines uncertain cues using readable rules. We check that raising risk inputs does not lower the output.
Does it prevent accidents?	No. It is an experimental cue for human review and does not replace existing safety controls.
Will it work at the event?	That is not established. Public training images and the eventual camera scene may differ.

Project location

The code, README, build guide, progress notes, and reports are in github.com/Cxen77/safesight. Before presenting, re-open the progress report and check if training or evaluation has changed.

Appendix: where the details live

Repository item	Purpose
README.md	Setup, dataset steps, run commands, dashboard, risk rules, and credits.
BUILD-GUIDE.md	Original phase-by-phase plan and acceptance gates.
PROGRESS.md	Recorded phase status and evidence. Some logs/status may be stale; verify current state.
src/	Inference, dataset tools, training, zones, fuzzy engine, pipeline, dashboard, evaluation.
docs/risk_rules.md	Readable 27-rule and 9-rule audit, overrides, and monotonic-calibration explanation.
results/	Preparation reports, verification logs, dashboard screenshots, and training status.
scenarios/template.csv	Header for future expert scenarios; no approved scenario cases yet.
checkpoints/ppe-baseline-resume.pt	Partial training resume point committed to GitHub; not a final inference model.

Upstream sources and terms

- SH17 PPE dataset and weights - CC BY-NC-SA 4.0 terms apply to derivatives.
- D-Fire dataset source - check the upstream license before redistribution.
- PPE-Detection-Jetson - upstream code attribution and GPL-3.0 notices are tracked in the repository.

Prepared from the SafeSight build guide, repository README, progress report, risk-rule audit, and dataset-preparation records. Document date: 05 November 2026.